

Key Punch Data:		County	Source	Emission	Action	Date	Agency/
		Number	Number	Point #	Type	Mo Day Yr	Indiv. ID
1	01101	0220	000003	0000	12	01/18/78	0001
1	3	4	7	8	12	13	15
16	18	52	54	59	66	68	80

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

010151

11

10

16 18

78

80

(NOTE: Columns not shown must be left blank)

(Do not punch Line 2 if column 78 is blank)

Company: Martin Marietta Cement Plant

SIC: _____

Address: LyonsCounty: BoulderPerson Contacted: Clarence Bureson, Gerald Anderson, Bob Reigel

Phone: _____

Inspected by: Parsons / Mesch / MironTime: 9:30 AM/PM

1. Insp to check new plan status and semi-annual update, as well as familiarization tour and EMS updating/clean up.
 2. Discussed new plans, reviewed calculations, plans, time tables. All unchanged since last meeting. Bill Miron said he felt a permit application was only needed for the dryer modification, other work on kiln and A-frame could be handled with ACEU update. This agrees with the information we have given M-M.
 3. Toured plant. Observed all processes and equipment. Saw where kiln modifications would be made.
- Process description and analysis. - see diagram on attached sheet
4. Quarry
 5. Primary crusher - receives shale. 600 TPH mill reduces material to minus 4" size. Byhouse on mill and conveyor.
 6. Surge silo and hopper. Byhouse on top.
 7. Dryer - Three byhouses plus stack
 8. Secondary crusher on silo. Byhouse on silo and mill. Unit is a 150 TPH hammer mill. Cuts to minus 1/2".

8. Roaster - gas fired - designed to drive off some of the volatiles.
9. Cooler - water spray on outside cools material and gases. Some hot gas recirculation to roaster burner. Drops temp from 900+° to 550°F.
10. Raw silo collects roasted shale.
11. Baghouse collects gases from roaster and conveyers. Exhausts to main stack.
12. Baghouses - unit rated at 210 000 cfm. will be converted to kiln side. two on conveyers carrying material to Raw silo at north end of kiln. Conveyers are covered.
13. Entire roaster process will be eliminated in upcoming modifications.
14. Kiln feed silos and homo silos - located next to roaster. Mix feed. Silo baghouse on top.
15. Coal handling - rail delivery currently from Wyoming. with upcoming modifications. new coal contract calls for west Slope coal from Massey Coal Co. Is reported to be lower sulfur content.
Rail car unload to conveyor (covered) into 4000 ton silo. 15 ton per hour hammer mill is in bottom of the silo. Feed directly into the kiln.
16. Kiln - coal fired
17. Control Room - mostly automatic control.

ACTION TYPE - DESCRIPTION

- 12 - Inspection
- 22 - Source Test
- 23 - Opacity Observation
- 24 - Odor Observation

COMPLIANCE STATUS CODES

- 0 - Unknown
- 1 - Not in Compliance
- 2 - In Compliance by Source Test
- 3 - In Compliance by Inspection
- 4 - In Compliance by Certification
- 5 - In Compliance with Increments of Progress
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- 7 - Unknown Compliance with Increments of Progress
- 8 - No Applicable State Regulation
- 9 - In Compliance -- Closed Down

17 (cont)

In control room: Alarm lights for "excess density" out of stack. Only roaster branching has bolometer. Auto alarm on unit lights control panel when emission exceeds 20% opacity. Bolometer reported as unreliable. Branching on kiln side has opening for bolometer but none is present. Mr Anderson said they plan to get both online and reliable after the upcoming modifications.

18. Kiln and plant can produce ~50 ~~52~~ tons per hour of cement. That is the rated capacity. Usual rate run is 49.5 tph. Rated annual output is 434 000 tons, best has been 403 000 tons. Usual is 390 tons per year of cement.

Kiln runs clinkers to A frame for storage.

19. Finishing mill and raw mill located receive clinkers and produce cement. Finish mill has a 4500 hp electric motor. Raw mill has a 3500 hp motor. Both are located north of the control room.

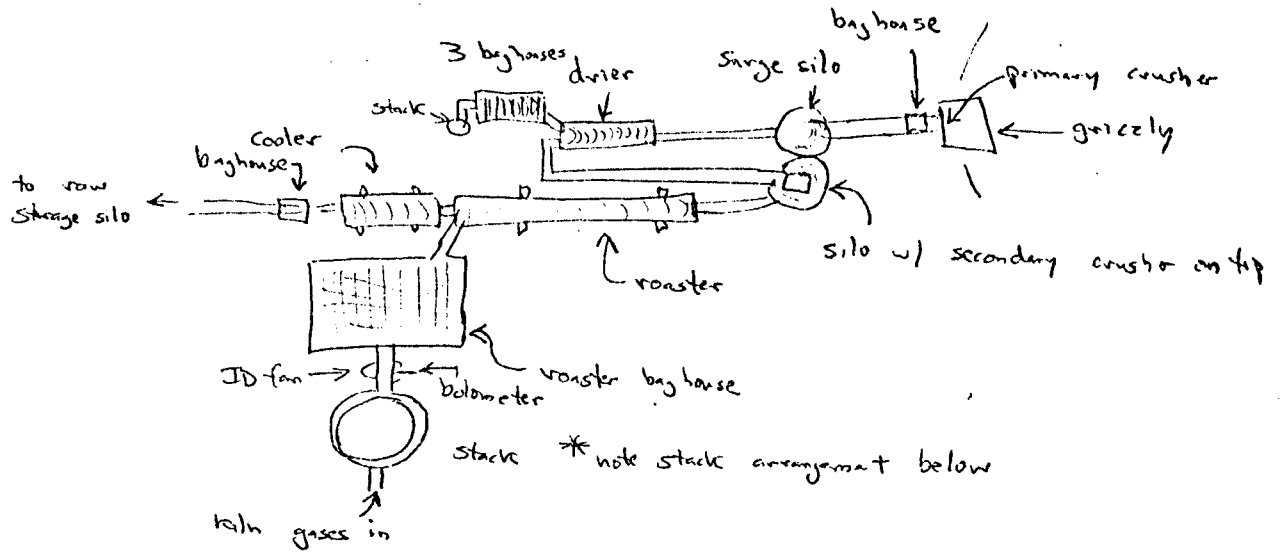
20. Product delivery area - truck unloading. Byhouse on silo roof. Unloading is under the silos.

21. Some wind area dust noted to Mr Anderson. He said they operate three vacuum cleaners and a water truck. Water rig seen later.

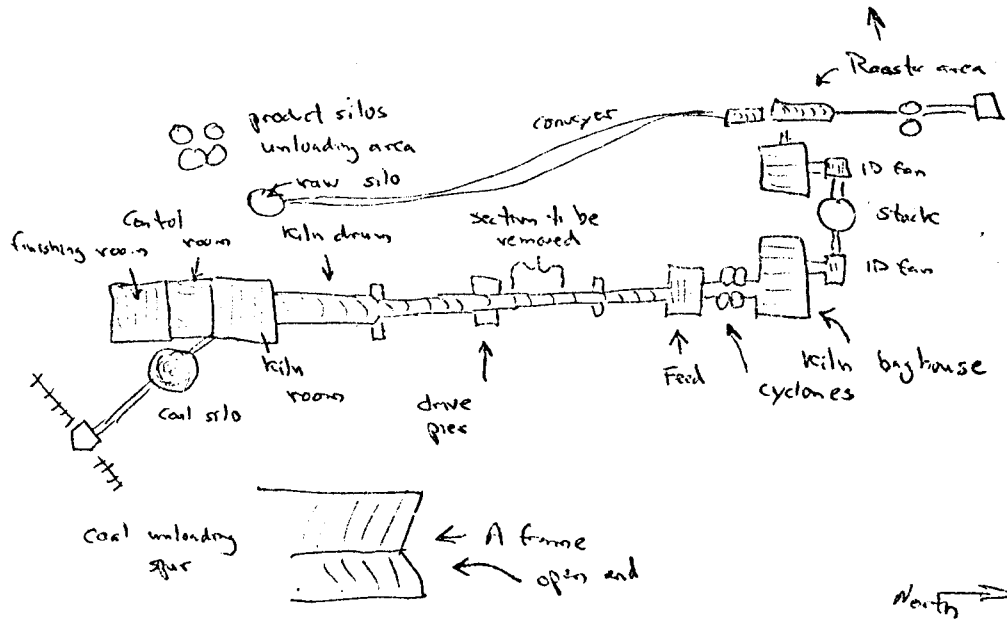
22. Excess emissions observed. Kiln coal feed valve (star valve) reported stuck, flamed out, air flow disrupted. Repaired within three minutes and stack cleared up.

(over for diagram)

Roaster process:



Kiln:



Key Punch Data:	County		Source		Emission		Action		Date			Agency/				
	Number		Number		Point #		Type		Mo	Day	Yr	Indiv. ID				
Line 1	101101	0220	00003	000	191971	12	041879	A116								
	1	3	4	7	8	12	13	15	16	18	52	54	59	66	68	80

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

101051	1	1	1
16	18	78	80

(NOTE: Columns not shown must be left blank)

(Do not punch Line 2 if column 78 is blank)

Company: Martin-Marietta SIC: _____

Address: P.O. Box 467 Lyons, Colo. County: Boulder

Person Contacted: Clarence Burleso, Gerald Anderson, Bob Riegel Phone: _____

Inspected by: Bill Miron, Mark Parsons, Ken Mesch Time: 9:45 AM/PM

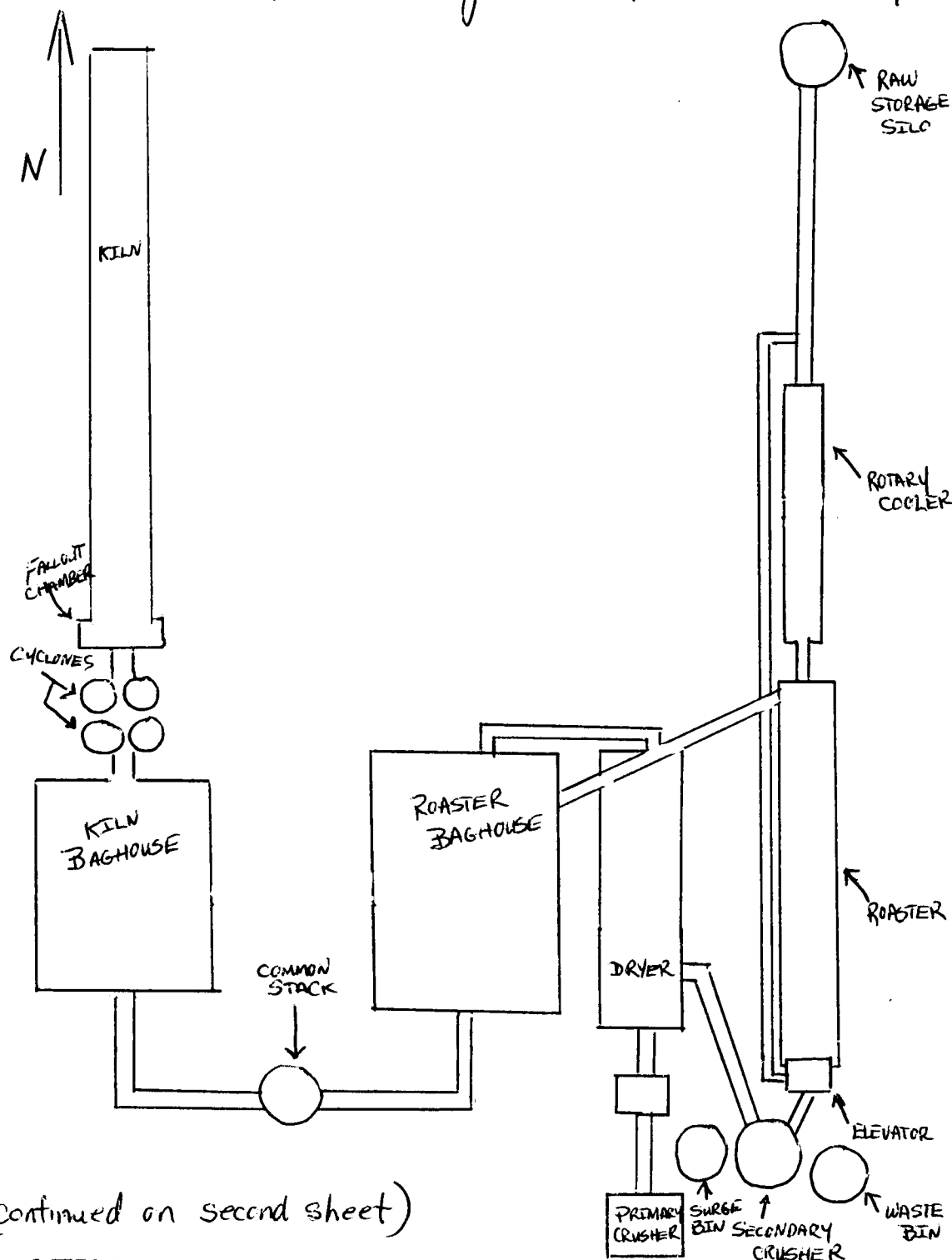
Martin-Marietta has plans (to be submitted soon) which will both reduce emissions and energy consumption at this cement manufacturing plant. These changes include:

- 1.) A bypass line from kiln to the existing kiln baghouse for impurities (this line will be appx. 900°F)
- 2.) A combustion chamber and spraytower (750°F) from which emission are to be routed to the existing roaster baghouse.
- 3.) The roaster/dryer will be eliminated.
- 4.) The kiln will be shortened
- 5.) The "A-frame" will be totally enclosed and moved to the south somewhat.

A tentative approximate schedule is as follows:

- 1.) "A-frame" - Engineering working on this
- 2.) May 7, 1979 - Construction to commence (A-frame)
- 3.) October 15, 1979 - Construction complete (A-frame)
- 4.) Fuller project
 - a.) July 1, 1979 - Construction commences
 - b.) January 1980 - tie in
 - c.) February 1980 - dry run
 - d.) March 1980 - start-up

General plant configuration prior to changes follows:



(continued on second sheet)

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AIR POLLUTION CONTROL - FIELD INSPECTION WORK SHEET

Key Punch Data:		County	Source	Emission	Action	Date	Agency/
		Number	Number	Point #	Type	Mo Day Yr	Indiv. ID
Line 1	101101	0220	00003	000	12	041879	A116
	1 3 4 7 8 12 13 15 16 18 52					54 59 66 68 80	

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

101051	11	101
16 18 78		80

(NOTE: Columns not shown must be left blank)

(Do not punch Line 2 if column 78 is blank)

Company: Martin-Marietta SIC: _____
 Address: Lyons, Colo County: Boulder
 Person Contacted: Gerald Anderson Phone: _____
 Inspected by: Bill Miron, Mark Parsons, Ken Mesch Time: 9:45 AM/PM

Dusting from the clinker storage pile in the "A-frame" building was occurring during the inspection (appx. 30-35%). A blue haze was coming from the common stack at appx 35-40% for a period of time. A check of operating conditions was made by Gerald Anderson and the flame had been lost in the roaster. This situation was corrected. The coal utilized at this plant comes from the west slope (Massy Coal Co.) and is stored in a 4000 Ton silo. The coal mill grinds the coal at 15 tons/hour and the coal is then blown into the kiln.

Point 001- Cement Kiln

SO₂ calculations: Given: ① production rate 1736 lb/min.
 ② coal rate 331 lb/min.

Sulfur balance:

IN { raw materials = 3492 lb/min × .0155 = 54.12 lb SO₃/min.
 Coal = 331 lb/min × .0088 × $\frac{80}{32}$ = 7.28 lb SO₃/min
 Total 61.40 lb SO₃/min. ③ 17T/hour collected particulate with 6.5% SO₃
 ④ kiln feed rate 3492 lb/min.
 ⑤ SO₃ molecular wt = 80
 ⑥ SO₂ molecular wt = 64

OUT { Clinkers 1736 lb/min × .0098 = 17.01 lb SO₃/min. ⑦ SO₃% in raw materials 1.55%
 Dust 17T/Hr × $\frac{2000 \text{ lb/T}}{60 \text{ min/Hr}} \times .065 = 36.83 \text{ lb SO}_3/\text{min}$ ⑧ SO₃% in Clinkers .98%

Total 53.84 lb SO₃/min

$$\text{SO}_3 \text{ lost} = 61.40 \text{ lb/min} - 53.84 \text{ lb/min} = 7.56 \text{ lb SO}_3/\text{min}$$

(over) $7.56 \text{ lb SO}_3/\text{min} \times \frac{64}{80} = 6.04 \text{ lb SO}_2/\text{min}$
 $\frac{6.04 \text{ lb SO}_2/\text{min}}{3492 + 331/2000} = 3.16 \text{ lb SO}_2/\text{Ton}$

$$\text{Total Tons of materials/year processed} = \frac{3492 \text{ lb/min} + 331 \text{ lb/min}}{1736 \text{ lb/min}} \times 403,000 \text{ T/yr.} = 887,482 \text{ T/yr.}$$

$$\text{Tons of SO}_2/\text{year} = \frac{3.16 \text{ lb/Ton} \times 887,482}{2000 \text{ lb/Ton}} = \underline{1402 \text{ Tons SO}_2/\text{year}}$$

Allowable SO₂ emissions as per Regulation No 1 section III, A.4.f.

7 lb SO₂/Ton of material processed including fuel

$$\frac{7 \text{ lb/Ton} \times 887,482 \text{ T/yr}}{2000 \text{ lb/Ton}} = 3106 \text{ T/year}$$

IN COMPLIANCE WITH REG. 1 section III, A.4.f.

Particulate emissions allowable as per Regulation No 1, section II, C.
(taken from figure 2)

$$E = 17.31 \times P^{0.16}$$

$$E = 17.31 \times \left(\frac{3492 + 331 \times 60}{2000} \right)^{0.16} = 36.97 \text{ /hour}$$

$$\frac{\text{lb/hr} \times 24 \text{ hr/day} \times \text{day/yr}}{2000 \text{ lb/Ton}} = 161.92 \text{ Tons/year}$$

Particulates (actual)

Tons of materials/year processed (including fuel) = 887,482 T/y
from AP-42, table 8.6-1

$$\frac{245 \text{ lbs/Ton} \times 887,482 \text{ T/yr}}{2000 \text{ lb/T}} \times .001 = \underline{108.71 \text{ Tons/year}}$$

In compliance with Reg. No. 1 section II c.

Visible emissions observation of the stack (with the exception of the previously mentioned loss of flame in Roaster) showed no violation of Regulation No. 1 section I.A.1.

(continued on page #3)

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Line 1	101101	0220	00003	000	12	041879	A16
	1 3 4	7 8	12 13 15	16 18	52	54 59	66 68 69

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

101051	1	101
16 18	78	80

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin Marietta SIC: _____
 Address: P.O. Box 467 Lyons, Colo County: Boulder
 Person Contacted: Gerald Anderson Phone: _____
 Inspected by: Bill Miner, Mark Parsons, Ken Mesch Time: 9:45 (AM/PM)

Point 001 (continued)

NO_x emissions from table 8.6-1 of AP-42

$$\frac{2.6 \text{ lbs/ton} \times 65,000 \text{ Tons/yr}}{2000 \text{ lbs/ton}} = 84.5 \text{ Tons/year}$$

The Federal primary and secondary standards for the ambient air are 100 µg/m³ for Nitrogen Dioxide (NO₂) on an annual average. No permit condition regarding NO_x emissions exists for this facility.

Point 002 Primary and Secondary Crushers

Both units were observed for visible emissions and none were noted from either source.

Primary crushing (actual) emissions

from table 8.20-1 AP-42

$$\frac{0.5 \text{ lb/Ton} \times 923,000 \text{ Tons/yr}}{2000 \text{ lbs/ton}} \times 0.001 = \underline{.23 \text{ Tons/year}}$$

Secondary crushing (actual) emissions

from table 8.20-1 AP-42

$$\frac{1.5 \text{ lb/Ton} \times 923,000 \text{ T/yr}}{2000 \text{ lb/Ton}} \times 0.001 = \underline{.69 \text{ Tons/year}}$$

allowable emissions in primary crushing as per Regulation No. 1 section II.C.

$$E = 17.31 \times P^{0.16}$$

$$E = 17.31 \times (200)^{0.16} = 40.4$$

$$E = 40.4 \text{ lbs/hour}$$

$$\frac{40.4 \text{ lbs/hr} \times 24 \text{ hrs/day} \times 7 \text{ days/week} \times 47 \text{ weeks/yr.}}{2000 \text{ lbs/Ton}} = 159.49 \text{ T/yr.}$$

secondary crushing allowable same as primary.
Both primary and secondary crushing are in compliance with Regulation No. 1 section II.C.

Point 003 Conveying

actual emissions from Table 8.20-1 AP-42

$$\frac{2 \text{ lbs/Ton} \times 900,000 \text{ Tons/year}}{2000 \text{ lbs/Ton}} \times 0.002 = \underline{1.8 \text{ Tons/year}}$$

allowable emissions from Reg. No. 1, section II.C.

$$E = 17.31 \times P^{0.16}$$

$$E = 17.31 \times (130)^{0.16}$$

$$E = 37.7 \text{ lbs/hour}$$

$$\frac{37.7 \text{ lbs/hour} \times 24 \text{ hrs/day} \times 7 \text{ days/wk} \times 47 \text{ wks/yr}}{2000 \text{ lbs/Ton}} = 148.83 \text{ T/yr}$$

conveying emissions in compliance with Reg. No. 1 section II.C.
No visible emission violation of Reg. 1, I.A.1 noted.

ACTION TYPE - DESCRIPTION

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AIR EMISSION CONTROL - FIELD INSPECTION WORK SHEET

Key Punch Data:	County	Source	Emission	Action	Date	Agency/
	Number	Number	Point #	Type	Mo Day Yr	Indiv. ID
Line 1	101101	0220	00003	000	191971	12
	1 3 4 7 8 12 13 15 16 18 52				041879	1116

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

00151	11	101
16 18 78		80

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin-Marietta

SIC: _____

Address: Lyons, Colo.County: BoulderPerson Contacted: Gerald Anderson

Phone: _____

Inspected by: Bill Miron, Mark Parsons, Ken MeschTime: 9:45 (AM) PM

Point 004 "A-frame" storage

was observed dusting at 30 to 35% opacity but was not leaving the property line. Emissions were not from any permanent stack, pipe or conduit. No analytical report of visible emission violation was written.

Allowable emissions from Reg. I section II.C.

$$E = 17.31 \times (50)^{0.16}$$

$$E = 32.36 \text{ lb/hour}$$

$$\frac{32.36 \text{ lb/hour} \times 24 \text{ hrs/day} \times 7 \text{ day/wk} \times 47 \text{ wk/yr}}{2000 \text{ lb/Ton}} = 143.54 \text{ T/yr.}$$

Actual emissions

$$.007 \times 50 \times 24 \times 7 \times 47 \times .003 = \underline{8.29 \text{ Tons/year}}$$

In compliance with Reg. M. 1 section II.C.

Point 006 Coal unloading & storage

allowable emissions from Reg. 1 section II.C.

$$E = 17.31 \times (200)^{0.16}$$

$$E = 40.4 \text{ lb/hour}$$

$$\frac{40.4 \times 1610 \text{ hrs/yr}}{2000 \text{ lb/T}} = 32.52 \text{ Tons/year}$$

point 006 (cont.)

actual emissions from silo :

$$.0002 \times 200 \text{ T/hour} \times 1610 \text{ hr/yr.} \times .0002 = \underline{\underline{.012 \text{ T/year}}}$$

actual emissions from dumping :

$$.007 \times 200 \text{ T/hr.} \times 1610 \text{ hr/yr.} \times .0002 = \underline{\underline{.45 \text{ tons/year}}}$$

In compliance with Reg. No. 1 section II.c. TOTAL 0.462 tons/year
No visible emissions noted, in compliance with Reg. 1, sect. IA1

Point 007 Slate pit quarry

No visible emissions noted off of the quarry site. In compliance with Reg. 1, section section II D.

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AIR

ION CONTROL - FIELD INSPEC

WORK SHEET

y	Punch Data:	County	Source	Emission	Action	Date			Agency/ Indiv. ID
		Number	Number	Point #		Type	Mo	Day	
1	101101	101101	101101	101101	191971	1121	104181791	111161	11
		1	3	6			54		50
		7	8	12	13	15		66	68
					16	18			80

Compliance Status

Line 2 (Dup. cols. 1-15 from Line 1)

101051	11	101
16	18	78
		80

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Martin Marietta

SIC: _____

Address: P.O. Box 447, Aurora, Colo.County: JeffersonPerson Contacted: George Peterson, Regional Manager

Phone: _____

Inspected by: Bill Miller, Mark Peterson, Ken MillerTime: 0745 AM/PM

EMISSIONS

POINT NUMBER	POLLUTANT	ACTUAL	ALLOWABLE	APPLICABLE REGULATION	COMPLIANCE STATUS
001	PM10	0.0000	0.0000	101, 111.0.	The compliance level for PM10 is 0.0000. The actual level is 0.0000. The compliance status is "COMPLIANT".
002	PM10	0.0000	0.0000	101, 111.0.	The compliance level for PM10 is 0.0000. The actual level is 0.0000. The compliance status is "COMPLIANT".
003	PM10	0.0000	0.0000	101, 111.0.	The compliance level for PM10 is 0.0000. The actual level is 0.0000. The compliance status is "COMPLIANT".

POINT 002 (continued)

PARTICULATES:

SECONDARY A (from AP-42, 8.20-1)

$$\frac{1.5 \text{ lb/T} \times 23,000 \text{ T/yr}}{2000 \text{ lb/T}} \times .001 = \underline{\underline{0.69 \text{ T/yr}}}$$

ALLOWABLE EMISSIONS (for both primary & secondary crushing from Reg. 1)

$$E = 17.31 \times P^{0.16}$$

$$E = 17.31 \times (200)^{0.16}$$

$$E = 40.4 \text{ lb/hr.}$$

$$\frac{40.4 \times 24 \times 7 \times 47}{2000} = \underline{\underline{159.5 \text{ T/yr}}}$$

POINT 003

PARTICULATES:

ACTUAL (from AP-42, 8.20-1)

$$\frac{2 \text{ lb/T} \times 500,000 \text{ T/yr}}{2000 \text{ lb/T}} \times 0.002 = \underline{\underline{1.8 \text{ T/yr}}}$$

ALLOWABLE (from Reg. 1, II.C.)

$$E = 17.31 \times P^{0.16}$$

$$E = 17.31 \times (130)^{0.16}$$

$$E = 37.7 \text{ lb/hr.}$$

$$\frac{37.7 \times 24 \times 7 \times 47}{2000} = \underline{\underline{148.8 \text{ T/yr}}}$$

POINT 004

PARTICULATES:

$$\text{ACTUAL} = .007 \times 50 \times 24 \times 7 \times 47 \times .003 = \underline{\underline{8.29 \text{ T/yr}}}$$

ALLOWABLE (from Reg. 1, II.C.)

$$E = 17.31 \times (50)^{0.16}$$

$$E = 32.36 \text{ lb/hr.}$$

$$\frac{32.36 \times 24 \times 7 \times 47}{2000} = \underline{\underline{143.5 \text{ T/yr}}}$$

POINT 006

PARTICULATES:

ACTUAL (from Silo)

$$.0002 \times 200 \text{ T/hr} \times 1610 \text{ hr/yr} \times .0002 = \underline{\underline{0.012 \text{ T/yr}}}$$

(from dumping)

$$.007 \times 200 \text{ T/hr} \times 1610 \text{ hr/yr} \times .0002 = \underline{\underline{0.45 \text{ T/yr}}}$$

$$\text{TOTAL} = \underline{\underline{0.462 \text{ T/yr}}}$$

ALLOWABLE (from Reg. 1, II.C.)

$$E = 17.31 \times (200)^{0.16}$$

$$E = 40.4 \text{ lb/hr}$$

$$\frac{40.4 \times 1610}{2000} = \underline{\underline{32.52 \text{ T/yr}}}$$

Compliance Status .

80

(Do not punch Line 2 if column 78 is blank)

AM/PM

- JUL 13 1979